



University of Asia Pacific

Department of Computer Science and
Engineering **CSE 437: Robotics Lab**

LAB REPORT

Experiment Number: 01

Experiment Title: Design and Implementation of a Line Following Robot.

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1. Experiment Name

Design and Implementation of a Line Following Robot

2. Objective

The objective of this experiment is to design and implement a line following robot using an Arduino microcontroller and infrared (IR) sensor module which can automatically follow a predefined path.

3. Apparatus / Hardware & Software Requirements

Hardware

- Arduino Uno
- IR Sensor Module (2 pieces – left and right sensor)
- DC Gear Motors (2 pieces)
- Motor Driver Module (L298N)
- Robot chassis with wheels
- Jumper wires
- Power supply (Battery pack)
- Breadboard

Software

- Arduino IDE
- Tinkercad / Wokwi for simulation

4. Circuit Diagram / Schematic

1) IR Sensor Connections

Five IR sensors are used to detect the position of the line.

Sensor Position Arduino Pin

Leftmost (S1) D2

Left (S2) D3

Center (S3) D4

Right (S4) D5

Rightmost (S5) D6

Common connections for all sensors:

- VCC → Arduino 5V
- GND → Arduino GND
- OUT → Corresponding digital pin

2) Motor Driver Connections (L293D)

Left Motor (Motor A)

L293D Pin Function Arduino Pin ENA Speed control (PWM)

D9

IN1 Direction control D8

IN2 Direction control D7

Right Motor (Motor B)

L293D Pin Function Arduino Pin ENB Speed control (PWM)

D10

IN3 Direction control D12

IN4 Direction control D11

3) Power Connections

Component Connection

L293D Vcc1 (Pin 16) Arduino 5V

L293D Vcc2 (Pin 8) 9V Battery (+)

L293D GND Pins (4,5,12,13) Common Ground Arduino

GND Battery GND

Note: All grounds must be connected together to form a common ground.

4) Summary of Arduino Pin Usage

Arduino Pins Purpose

D2–D6 IR Sensors

D7–D8 Left motor direction D9 Left motor speed (PWM) D10 Right motor speed (PWM) D11–D12 Right motor direction

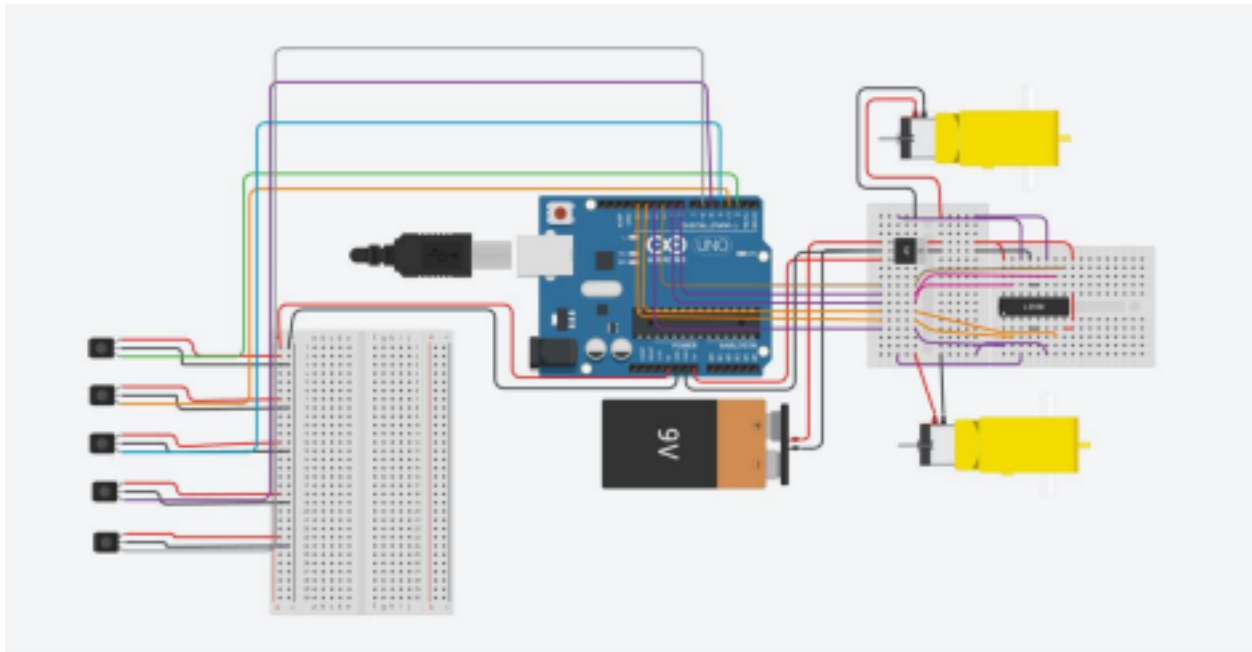


Fig 4.1: Circuit Diagram for Line Following Robot

5. Code / Assembly Program

// Sensor pins

int S1 = 2; // leftmost

int S2 = 3; // left

int S3 = 4; // center

int S4 = 5; // right

int S5 = 6; // rightmost

// Motor A (left motor)

int ENA = 9;

int IN1 = 8;

```
int IN2 = 7;

// Motor B (right motor)

int ENB = 10;
```

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```
int IN3 = 12;

int IN4 = 11;

int speedA = 150;

int speedB = 150;

void setup() {

  pinMode(S1, INPUT);

  pinMode(S2, INPUT);

  pinMode(S3, INPUT);

  pinMode(S4, INPUT);

  pinMode(S5, INPUT);

  pinMode(ENA, OUTPUT);

  pinMode(IN1, OUTPUT);

  pinMode(IN2, OUTPUT);

  pinMode(ENB, OUTPUT);

  pinMode(IN3, OUTPUT);

  pinMode(IN4, OUTPUT);

}

void loop() {

  int s1 = digitalRead(S1); int s2

= digitalRead(S2); int s3 =
```

```
digitalRead(S3);    int  s4  =  
digitalRead(S4);    int  s5  =  
digitalRead(S5); // Center on  
line → go forward  
if (s3 == LOW) {
```

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```
    forward();  
}  
// Line toward left  
else if (s2 == LOW) {  
    turnLeft();  
}  
else if (s1 == LOW) {  
    sharpLeft();  
}  
// Line toward right  
else if (s4 == LOW) {  
    turnRight();  
}  
else if (s5 == LOW) {  
    sharpRight();  
}  
// No line detected  
else {  
    stopMotors();
```

```
}
```

```
}
```

```
// Movement functions
```

```
void forward() {
```

```
  analogWrite(ENA, speedA);
```

```
  analogWrite(ENB, speedB);
```

```
  digitalWrite(IN1, HIGH);
```

```
  digitalWrite(IN2, LOW);
```

```
  digitalWrite(IN3, HIGH);
```

```
  digitalWrite(IN4, LOW);
```

```
}
```

```
void turnLeft() {
```

```
  analogWrite(ENA, 80);
```

```
  analogWrite(ENB, speedB);
```

```
  digitalWrite(IN1, HIGH);
```

```
  digitalWrite(IN2, LOW);
```

```
  digitalWrite(IN3, HIGH);
```

```
  digitalWrite(IN4, LOW); }
```

```
void sharpLeft() {
```

```
  analogWrite(ENA, 0);
```

```
  analogWrite(ENB, speedB);
```

```
  digitalWrite(IN1, LOW);
```

```
  digitalWrite(IN2, LOW);
```

```
  digitalWrite(IN3, HIGH);
```

```
digitalWrite(IN4, LOW); }  
  
void turnRight() {  
  analogWrite(ENA, speedA);  
  analogWrite(ENB, 80);  
  digitalWrite(IN1, HIGH);
```

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```
  digitalWrite(IN2, LOW);  
  digitalWrite(IN3, HIGH);  
  digitalWrite(IN4, LOW);  
}  
  
void sharpRight() {  
  analogWrite(ENA, speedA);  
  analogWrite(ENB, 0);  
  digitalWrite(IN1, HIGH);  
  digitalWrite(IN2, LOW);  
  digitalWrite(IN3, LOW);  
  digitalWrite(IN4, LOW);  
}  
  
void stopMotors() {  
  analogWrite(ENA, 0);  
  analogWrite(ENB, 0);  
}
```

6. Output / Observations

After uploading the program to the Arduino board and placing the robot on a black line track, the robot successfully followed the line.

Observation:

- When both sensors detect the black line, the robot moves forward.
- When the left sensor detects the line and the right sensor does not, the robot turns left. •
- When the right sensor detects the line and the left sensor does not, the robot turns right. •
- When both sensors fail to detect the line, the robot stops.

The robot was able to follow simple straight and curved paths.

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7. Result

The line following robot was successfully designed and implemented. The robot followed the given path correctly using IR sensors and Arduino. Therefore, the objective of this experiment was achieved.

8. Conclusion

In this experiment, a line following robot was designed using Arduino, IR sensors and a motor driver module. From this project, we learned how sensors can be used for environment detection and how a microcontroller can control motors based on sensor input. This experiment also helped to improve understanding of embedded systems and basic robotics control.

